# Build and Deploy Selenium Grid for Chrome and Firefox

## Step 1: Launch AWS EC2 Instance

- Go to AWS EC2 Console
- Launch a new instance with the following settings:
  - **Name:** SeleniumGridServer
  - **AMI:** Ubuntu Server 22.04 LTS
  - **Instance Type:** t2.micro (for demo) or **t3.small (recommended)**
  - **Key Pair:** Create or select existing
  - **Security Group Rules:**
    1. SSH (22)
    2. Custom TCP – Port **4444** (Anywhere)

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## Step 2: Connect to EC2 Instance

- Connect using SSH

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## Step 3: Install Docker and Docker Compose

```
sudo apt update
sudo apt install -y docker.io
sudo systemctl start docker
sudo systemctl enable docker
```

### Install Docker Compose

```
sudo curl -
L"https://github.com/docker/compose/releases/download/v2.17.3/docker-compose-
$(uname-s)-$(uname-m)"-o /usr/local/bin/docker-compose


sudo chmod +x /usr/local/bin/docker-compose
```

Verify:

```
docker--version
docker-compose--version
```

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## Step 4: Create Selenium Grid with Docker Compose (Stable Version)

```
mkdir selenium-grid
cd selenium-grid

nano docker-compose.yml
```

Paste **this updated and stable configuration**:

```
version:"3.8"

services:
selenium-hub:
image: selenium/hub:4.18.1
container_name: selenium-hub
ports:
-"4444:4444"

chrome:
image: selenium/node-chrome:4.18.1
container_name: chrome-node
shm_size: 2gb
depends_on:
- selenium-hub
environment:
- SE_EVENT_BUS_HOST=selenium-hub
- SE_EVENT_BUS_PUBLISH_PORT=4442
- SE_EVENT_BUS_SUBSCRIBE_PORT=4443

firefox:
image: selenium/node-firefox:4.18.1
container_name: firefox-node
shm_size: 2gb
depends_on:
- selenium-hub
environment:
- SE_EVENT_BUS_HOST=selenium-hub
- SE_EVENT_BUS_PUBLISH_PORT=4442
- SE_EVENT_BUS_SUBSCRIBE_PORT=4443
```

Save and exit.

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## Step 5: Start Selenium Grid

```
sudo docker-compose down
sudo docker system prune -af
sudo docker-compose up -d
```

Verify containers:

```
docker ps
```

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## Step 6: Access Selenium Grid UI

Open in browser:

```
<EC2-PUBLIC-IP>:4444/ui
```

You should see **Chrome and Firefox nodes registered**.

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## Step 7: Run Sample Python Test (Optional)

### Install Python Requirements (Minimal)

```
sudo apt install -y python3
```

```
sudo apt install -y python3-pip
```

```
sudo apt install -y python3-venv
```

```
sudo apt update
```

### Create Virtual Environment

```
python3-m venv venv
source venv/bin/activate
```

### Install Latest Selenium

```
pip install --upgrade pip
pip install selenium
```

Verify version:

```
pip show selenium
```

(Should be **4.18.x or latest**)

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### Create Test Script

```
nano test_grid.py
```

Paste this **final working code**:

```
from selenium import webdriver

GRID_URL ="http://localhost:4444"

options = webdriver.ChromeOptions()
options.add_argument("--headless=new")
options.add_argument("--no-sandbox")
options.add_argument("--disable-dev-shm-usage")
options.add_argument("--disable-gpu")
options.add_argument("--window-size=1920,1080")

driver = webdriver.Remote(
    command_executor=GRID_URL,
    options=options
)

driver.get("https://www.google.com")
print("Title:", driver.title)

driver.quit()
```

Save and exit.

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## Execute Test

```
python3 test_grid.py
```

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